Date	01 October 2026	**Maximum Marks**	2 Marks
Team ID	SWTID-2026-5883	**Document Type**	Technology Stack Details
Project Name	**Pocket Smart AI: Your Smart Budget & Recommendation Assistant**		

TECHNOLOGY STACK SPECIFICATION

Define Your Technology Stack: Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Team Members & Assigned Architecture Component

S.No	Student Name	Assigned Stack Component	Responsibilities & Technical Ownership
1	**Jashvini.K**	1. Frontend / Client-Side Framework	Design cross-platform UI, user dashboards, interactive budget visualizations, and client-side responsive layout.
2	**Indhumathi.D**	2. Backend / Server-Side & AI Engine	Develop RESTful microservices, AI recommendation backend models, income-expense calculation, and JWT token management.
3	**Jayapriya Rajkumar**	3. Database & Data Storage Layer	Configure relational database schemas, secure transaction storage, cache layers, and historical spend queries.
4	**Jonal.A**	4. Cloud Infrastructure & Deployment	Manage containerized deployments, cloud hosting, auto-scaling groups, and load balancing setup.
5	**Kavibharathi.K**	5. Version Control, CI/CD & APIs	Setup automated build pipelines, code quality pipelines, Open Banking integrations, and push notification APIs.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** (React Native / Flutter / HTML5)	**Flutter (Dart) / React Native**	Enables single-codebase cross-platform mobile app development for iOS and Android, offering high-performance UI components, smooth budget chart animations, and seamless responsive design.
2	**Backend / Server-Side** (Node.js / Python Fast-API)	**Python (FastAPI & PyTorch)**	Provides rapid REST API execution speed, asynchronous request handling, and direct native integration with AI/ML libraries for budget prediction and smart recommendation algorithms.
3	**Database / Data Storage** (PostgreSQL / Redis)	**PostgreSQL & Redis Cache**	PostgreSQL offers ACID-compliant relational mapping for secure transaction logs and financial records, while Redis

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
			provides sub-millisecond in-memory caching for active user session budgets.
4	**Cloud / Hosting / Deployment** (AWS / Docker / Kubernetes)	**AWS (ECS, Lambda) & Docker**	Delivers cost-effective serverless execution for budget notifications, containerized microservices for high reliability, and automatic horizontal scaling during traffic peaks.
5	**Version Control & CI/CD** (GitHub / GitHub Actions)	**GitHub & GitHub Actions**	Facilitates seamless team code collaboration, pull-request code reviews, and automated unit testing, security scanning, and deployment pipelines.
6	**Third-Party APIs / Tools** (Plaid / Firebase FCM / OpenAI API)	**Plaid API, Firebase FCM & OpenAI API**	Plaid enables secure bank transaction synchronization, Firebase FCM dispatches real-time budget threshold push alerts, and OpenAI API provides intelligent personal financial advice.